

	Success	Efficiency			Interaction quality					
	In-time	Action	Conversation	Overall	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness	Overall
Assistant										
Frontier API models										
Gemini 3.8 Flash	21.4	31.1	84.6	58.2	<u>82.7</u>	83.0	67.6	62.9	79.0	46.2
Gemini 3.1 Pro	17.6	32.7	<u>87.5</u>	60.8	71.4	78.0	61.3	67.8	68.6	30.8
GPT-6 Astra	15.8	37.9	91.3	65.1	76.9	84.0	56.6	79.0	72.8	34.5
Real-time streaming API models										
GPT-Realtime 2.1	19.0	36.5	80.2	58.9	82.1	87.6	<u>66.6</u>	<u>77.4</u>	54.1	37.1
Gemini 3.8 Live	18.1	39.2	82.1	60.9	89.9	76.4	62.3	52.3	80.9	<u>42.1</u>
Open-weight non-streaming VLMs										
Qwen3.8-27B	25.2	32.9	85.6	58.7	68.7	78.7	62.8	67.2	58.5	28.5
Kimi-VL-A3B-Thinking	<u>24.3</u>	34.1	83.7	58.1	56.4	60.0	42.4	37.1	30.8	12.0
InternVideo3-8B	18.2	36.7	79.6	56.9	64.0	61.7	44.9	24.4	44.4	18.4
Muse Glimmer	15.7	39.2	75.7	58.6	70.3	<u>85.1</u>	62.6	75.9	45.2	26.4
Molmo2-8B	2.6	<u>40.9</u>	80.2	<u>65.0</u>	72.2	55.8	42.0	20.3	41.6	17.0
Open-weight streaming VLMs										
Proact-VL	20.2	37.9	63.6	50.0	69.7	30.1	22.0	20.5	25.5	16.3
StreamingVLM	19.0	29.8	81.5	54.7	76.7	34.4	29.4	29.0	19.5	23.4
MiniCPM-o 4.5	15.4	34.8	55.4	44.3	61.7	58.5	35.5	39.1	29.4	16.2
Qwen3-Omni-30B	13.4	45.9	65.5	55.1	75.7	83.1	60.6	59.5	<u>79.1</u>	30.2
Reference conditions										
No assistance	15.5	20.0	100.0	57.5	—	—	—	—	—	—

Table 2 Visual assistance in VHSim. All scores are on a 0–100 scale (↑). Success and efficiency are three-run means; efficiency is measured on in-time successes, with persona-weighted overall scores. Rubrics use run 1 and the full 16-item overall score. Bold/underline indicate the best/second-best assistant in each column; dashes denote unavailable scores.